De Morgan Laws

(i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$
(iii) $A - (B \cup C) = (A - B) \cap (A - C)$ (iv) $A - (B \cap C) = (A - B) \cup (A - C)$.

On Symmetric Difference

(i) $A \Delta B = B \Delta A$ (ii) $(A \Delta B) \Delta C = A \Delta (B \Delta C)$
(iii) $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$.

On Empty Set and Universal Set

(i) $\emptyset' = U$ (ii) $U' = \emptyset$
(iii) $A \cup A' = U$ (iv) $A \cap A' = \emptyset$
(v) $A \cup U = U$ (vi) $A \cap U = A$.

On Cardinality

(i) For any two finite sets A and B, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
(ii) If A and B are disjoint finite sets, then $n(A \cup B) = n(A) + n(B)$.
(iii) For any three finite sets A, B and C,
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$.

1.3 Cartesian Product

We know that the Cartesian product of sets is nothing but a set of ordered elements. In particular, Cartesian product of two sets is a set of ordered pairs, while the Cartesian product of three sets is a set of ordered triplets. Precisely, let A, B and C be three sets. Then the ***Cartesian product*** of A with B is denoted by $A \times B$. It is defined by

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

Similarly, the Cartesian product $A \times B \times C$ is defined by

$$A \times B \times C = \{(a, b, c) : a \in A, b \in B, c \in C\}.$$

Thus $A \times A = \{(a, b) : a, b \in A\}$.

Is it correct to say $A \times A = \{(a, a) : a \in A\}$?

It is important that the elements of the Cartesian product are ordered and hence, for non-empty sets,

$$A \times B \neq B \times A, \text{ unless } A = B.$$

That is, for non-empty sets $A \times B = B \times A$ if and only if $A = B$. We know that $\mathbb{R}$ denotes the set of real numbers and

$$\begin{aligned} \mathbb{R} \times \mathbb{R} &= \{(x, y) : x, y \in \mathbb{R}\}. \\ \mathbb{R} \times \mathbb{R} \times \mathbb{R} &= \{(x, y, z) : x, y, z \in \mathbb{R}\}. \end{aligned}$$

Symbolically, $\mathbb{R} \times \mathbb{R}$ can be represented as $\mathbb{R}^2$ and $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ as $\mathbb{R}^3$. Note that $\mathbb{R} \times \mathbb{R}$ is a set of ordered pairs and $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ is a set of ordered triplets.

If $A = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$ then

$$A \times B = \{(1, 2), (1, 4), (1, 6), (2, 2), (2, 4), (2, 6), (3, 2), (3, 4), (3, 6)\}.$$